

Computer Networks

Course Code: CS208

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Course Description

Unit No.	Description of Unit
Unit 1	Foundations of Computer Networks: Evolution of computer networks, basic principles of network design, network topologies (Star, Mesh, Bus, Ring, Hybrid), OSI and TCP/IP models, physical layer fundamentals including data and signals, bandwidth utilization, transmission techniques and media, signal encoding, and multiplexing.
Unit 2	Data Link and Network Layer Protocols: Error detection and correction mechanisms, data link control protocols, multiple access protocols and LAN technologies, network layer functions and services, routing algorithms (Link-State, Dijkstra's algorithm), IP addressing, subnetting, super netting (CIDR), IPv4, IPv6, and address resolution protocols such as ARP, RARP, BOOTP, and DHCP.
Unit 3	Advanced Networking and Routing Protocols: Advanced IP addressing, routing protocols including RIP, OSPF, and BGP, multicast routing and protocols, Quality of Service (QoS) concepts and protocols, and network security fundamentals.

Course Description

Unit No.	Description of Unit
Unit 4	Transport Layer and End-to-End Protocols: Transport layer services including UDP and TCP, flow control, congestion control, TCP congestion avoidance algorithms, session management (establishment and synchronization), and presentation layer functions such as data formatting, encryption, and decryption.
Unit 5	Application Layer Protocols and Network Management: Naming and addressing mechanisms: DNS, URIs; Communication and data transfer protocols: HTTP, SSH, Telnet, SMTP, POP3, IMAP, FTP, SFTP, FTPS; World Wide Web fundamentals; Network management with SNMP, Modern application-layer protocols: WebSockets, MQTT.

Recommended Reading Material

S. No.	Name of Books / Authors / Publishers
1	Behrouz A. Forouzan, Data Communications and Networking, 6th ed., McGraw Hill
2	William Stallings, Data and Computer Communications, 10th ed., Pearson Education
3	Andrew S. Tanenbaum, Computer Networks, 6th ed., Pearson Education India
4	J. F. Kurose, K. W. Ross, Computer Networking: A Top-Down Approach, 8th ed., Addison-Wesley

Course Outcomes(CO)

S. No	Course Outcomes (CO)
CO1	Understand the foundational aspects of computer networks, including network topologies, OSI and TCP/IP models, to design and analyze network structures effectively.
CO2	Master error detection and correction, data link control, and IP addressing, enabling efficient data communication across networks.
CO3	Learn advanced routing protocols, Quality of Service (QoS) principles, and basic network security to manage high-performance, secure networks.
CO4	Acquire a deep understanding of transport protocols like UDP and TCP, focusing on reliable data transfer and congestion control.
CO5	Develop expertise in application layer protocols and network management, preparing for effective management of networked applications.

Evaluation Criteria

Components	Maximum Marks
CWS	15 (Attendance, Assignments/Quiz, Class Performance)
PRS	25 (Practical)
MTE	20 (Mid Term)
ETE	40 (End Term)
Total	100

Google Classroom

- The study material, Assignment, Quiz, and all other instructions related to the Course will be posted on Google Classroom.
- All students are requested to join the **Computer Network, Section-A3** Google Classroom with the following Code:

winwekww